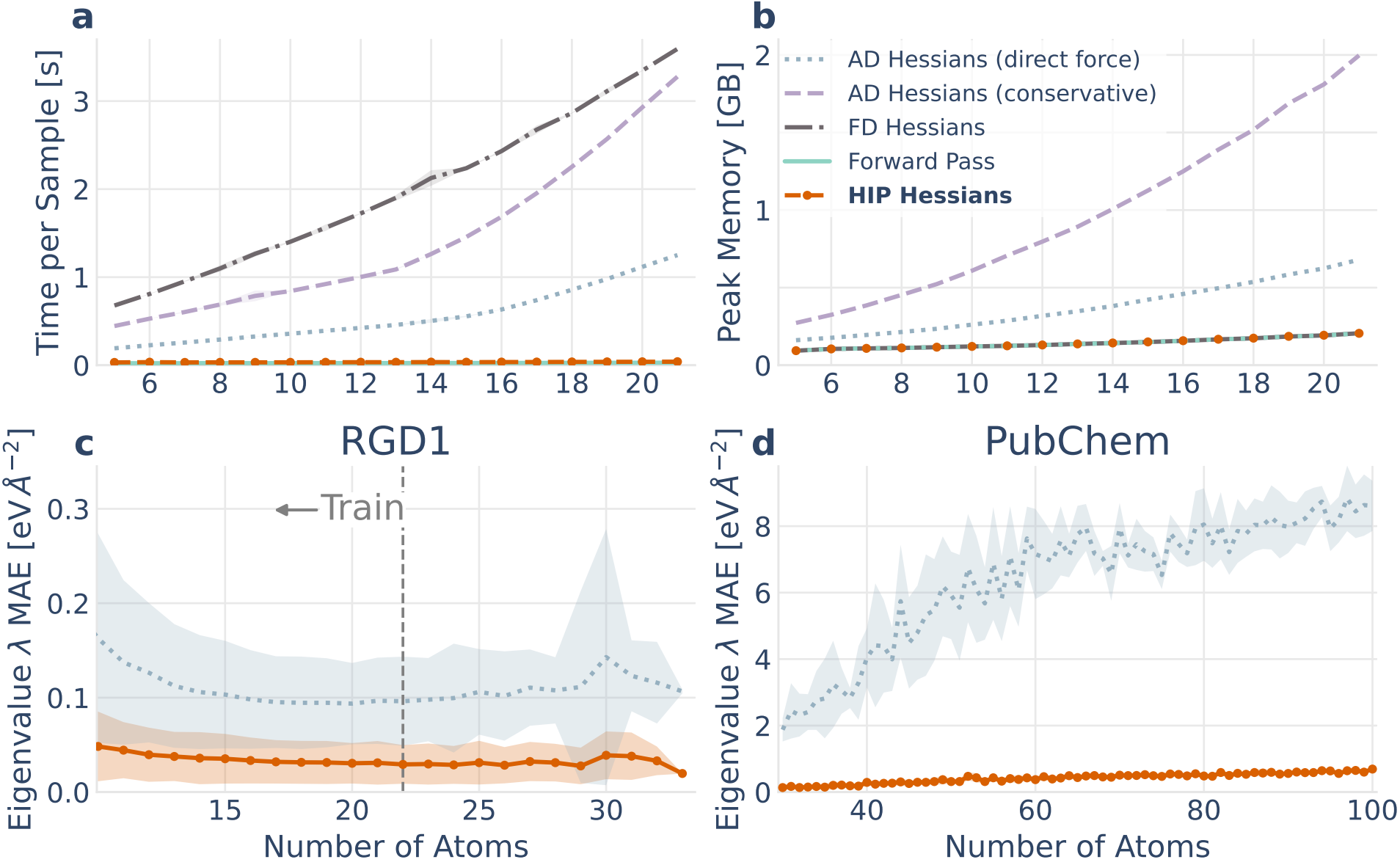
a
Time per Sample [s]
0
1
2
3
6
8
10
12
14
16
18
20
b
Peak Memory [GB]
0
1
2
AD Hessians (direct force)
AD Hessians (conservative)
FD Hessians
Forward Pass
HIP Hessians
6
8
10
12
14
16
18
20
c
RGD1
Train
Eigenvalue λ MAE [eV Å$^{-2}$]
0.0
0.1
0.2
0.3
15
20
25
30
Number of Atoms
d
PubChem
Eigenvalue λ MAE [eV Å$^{-2}$]
0
2
4
6
8
40
60
80
100
Number of Atoms